

The Confusion Matrix

How a Classifier Is Right and Wrong

AWS Machine Learning Associate • Part 17

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What We'll Cover



What it is & the four outcomes



A worked example



Metrics from the matrix



Which error matters more?

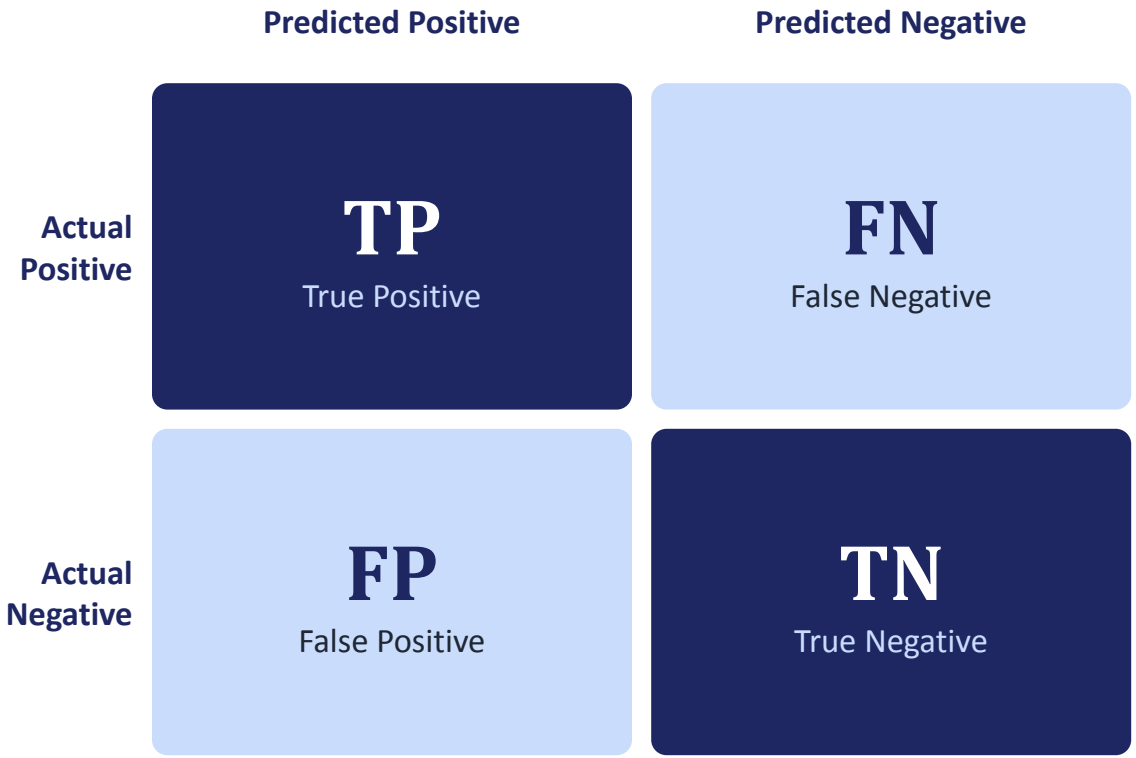


Thresholds & the ROC curve



Multi-class & SageMaker

What Is a Confusion Matrix?



How to read the names

- **2nd word**
  What the model predicted: positive or negative
- **1st word**
  Was that prediction right? True or false

Rows show reality, columns show the model's prediction. Some tools flip this, so check the labels.

The Four Outcomes (Fraud Example)



True Positive

Flagged as fraud, and it was fraud. A successful catch



True Negative

Said legitimate, and it was legitimate. The normal, quiet case



False Positive (Type I)

Flagged as fraud, but it was legitimate. A false alarm



False Negative (Type II)

Said legitimate, but it was fraud. A miss

A Worked Example

	Predicted Positive	Predicted Negative
Actual Positive	**8** True Positives	**2** False Negatives
Actual Negative	**4** False Positives	**986** True Negatives

1,000 transactions

- **10 actual frauds**
  The model caught 8 and missed 2
- **990 legitimate**
  The model passed 986 and flagged 4
- **12 flagged in total**
  8 of them were real fraud

Dark cells are correct predictions. Light cells are the errors.

Metrics from the Matrix

Metric	Formula	The question it answers	Our example
Accuracy	`(TP + TN) / Total`	How often is the model right overall?	**99.4%**
Precision	`TP / (TP + FP)`	When it says fraud, how often is it right?	**66.7%**
Recall (Sensitivity)	`TP / (TP + FN)`	Of all real fraud, how much did we catch?	**80%**
Specificity	`TN / (TN + FP)`	Of all legitimate cases, how many did we pass?	**99.6%**
F1 Score	`2 × P × R / (P + R)`	A balance of precision and recall	**72.7%**

The Accuracy Trap, Revisited

A Lazy Model

Always predicts "not fraud"

Accuracy	**99.0%**
Recall	**0%**
Precision	**undefined**

Our Model

Catches 8 of 10 frauds

Accuracy	**99.4%**
Recall	**80%**
Precision	**66.7%**

High accuracy can hide a useless model. The confusion matrix shows it.

Which Error Matters More?

Missing a real case is costly (disease screening, fraud)	→ **Optimize Recall**
False alarms are costly (a spam filter hiding real email)	→ **Optimize Precision**
You need a balance of both	→ **Use the F1 Score**
Classes are balanced and both errors cost about the same	→ **Accuracy is fine**

Thresholds & the ROC Curve

Lower the Threshold

Recall ↑ Precision ↓

Flags more cases: more catches, but more false alarms.

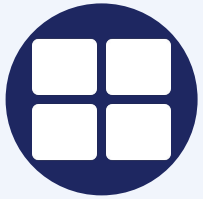
Raise the Threshold

Precision ↑ Recall ↓

Flags fewer cases: fewer false alarms, but more misses.

Each threshold gives a different confusion matrix. The ROC curve plots true positive rate vs. false positive rate across all thresholds, and AUC summarizes it in one number.

Beyond Two Classes & in SageMaker



Multi-Class Matrices

The grid grows to N × N, and the diagonal shows the correct predictions



Spot the Confusion

Off-diagonal cells show which classes the model mixes up



Normalize the Rows

Percentages per row make imbalanced classes easier to compare



In SageMaker

Look for these metrics in Autopilot model reports and Model Monitor model quality checks

Recap

- Four cells compare predictions with reality: TP, TN, FP, FN.
- From them: accuracy, precision, recall, specificity & F1.
- Match the metric to the cost of your errors, and never trust accuracy alone.

Thanks for watching!